

General Spatio-Temporal Factor Models for High-Dimensional Random Fields on a Lattice

(Barigozzi, La Vecchia & Liu)

Time Series Econometrics

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Introduction

- High-dimensional spatio-temporal random field on a lattice:

$$x_{n\varsigma} = (x_{1\varsigma}, \dots, x_{n\varsigma})^\top.$$

where

$$\varsigma = (s_1, s_2, t)^\top, \quad s_1 = 1, \dots, S_1, \quad s_2 = 1, \dots, S_2, \quad t = 1, \dots, T \quad \text{and} \quad \ell = 1, \dots, n.$$

- $\theta = (\theta_1, \theta_2, \theta_3)^\top \in [-\pi, \pi]^3$ denotes a spatio-temporal frequency.
- $h = (h_1, h_2, h_3)^\top \in \mathbb{Z}^3$ a spatio-temporal lag.

The General Spatio-Temporal Factor Model (GSTFM). Model A

Decomposition:

$$x_{\ell,s} = \chi_{\ell,s} + \xi_{\ell,s},$$

where $\chi_{\ell,s}$ - common component driven; and $\xi_{\ell,s}$ - idiosyncratic term.

Problem: We don't know which components are common.

GDFM vs Spatial Structure. Model B

- GDFM are designed for high-dimensional time series, but ignores spatial dependence.
- Stacking the space-time field into a vector $x_{N,t}$ with $N = nS_1S_2$ destroys spatial structure.
- As m increases, leading eigenvalues grow together and the eigen-gap weakens.

Research Questions

How can we preserve spatial dependence?

Can the problem "curse of dimensionality" be solved and how?

Solution

GSTFM and blessing of dimensionality

- under a spatio-temporal factor structure,

$$\bar{\lambda}_j(m) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \hat{\lambda}_{x,j}^{(m)}(\theta), \quad j = 1, \dots, q+1, \quad m - \text{cross-sectional dimension}$$

- the spectral density matrix of x exhibits exactly q diverging eigenvalues as $n \rightarrow \infty$,
- the remaining eigenvalues stay bounded,
- the **eigen-gap**. The widening gap between $\bar{\lambda}_q(m)$ and $\bar{\lambda}_{q+1}(m)$ provides empirical evidence for identification thr number of common factors.

GSTFM diagnostic

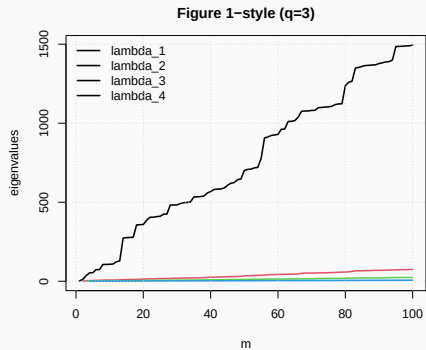
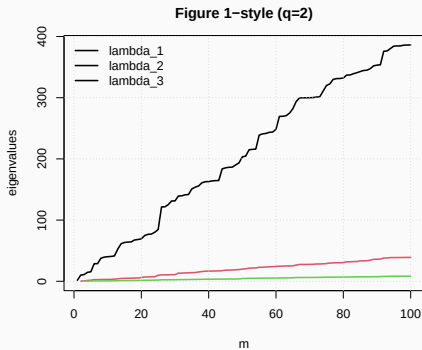


Figure 1: Figure 1 (GSTFM diagnostic). Averaged eigenvalues of the kernel-smoothed spatio-temporal spectral density estimator as the cross-sectional dimension m increases. Under GSTFM, the first q eigenvalues increase strongly while the $(q + 1)$ -th remains relatively stable, producing an eigen-gap that identifies q .

Stacked GDFM diagnostic

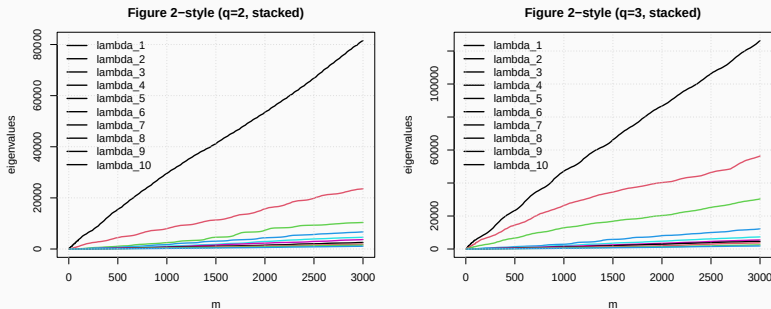


Figure 2: Figure 2 (stacked GDFM diagnostic). Averaged eigenvalues of the spectral density estimator for the stacked time series representation. The eigen-gap weakens or disappears as the dimension grows

Note: we stack the spatio-temporal field into a high-dimensional vector time series

$$Z \in \mathbb{R}^{T \times N}, \quad N = nS_1S_2,$$

Population model

Population model: Assumptions

- The q -GSTFM decomposition:

$$x_{\ell\varsigma} = \chi_{\ell\varsigma} + \xi_{\ell\varsigma}, \quad \ell = 1, \dots, n. \quad (1)$$

- The q common components ($q \ll n$) of latent spatio-temporal factors $u_{\varsigma} = (u_{1\varsigma}, \dots, u_{q\varsigma})^{\top}$
- Linear filter:

$$\chi_{\ell\varsigma} = b_{\ell}(L) u_{\varsigma} = \sum_{\kappa \in \mathbb{Z}^3} \sum_{j=1}^q b_{\ell j, \kappa} u_{j, \varsigma - \kappa}. \quad (2)$$

- $\xi_{\ell\varsigma}$ is weakly cross-sectionally correlated.

Population model: Spectral density matrix

- Spatio-temporal autocovariance matrix at lag h :

$$\Gamma_n^x(h) = \mathbb{E} \left[x_{n,\varsigma+h} x_{n\varsigma}^\top \right].$$

- Spatio-temporal spectral density matrix $n \times n$:

$$\Sigma_n^x(\theta) = \frac{1}{8\pi^3} \sum_{h \in \mathbb{Z}^3} \Gamma_n^x(h) e^{-i\langle h, \theta \rangle}. \quad (3)$$

- $\lambda_{n1}^x(\theta) \geq \dots \geq \lambda_{nn}^x(\theta)$ be the eigenvalues of $\Sigma_n^x(\theta)$.

Population model: Identification condition

The Identification of the GSTFM is the *eigen-gap* in the dynamic eigenvalues for almost every $\theta \in [-\pi, \pi)^3$,

$$\lambda_{nq}^x(\theta) \rightarrow \infty \quad \text{while} \quad \sup_n \lambda_{n,q+1}^x(\theta) < \infty.$$

- the first q eigenvalues correspond to the common dynamics,
- the remaining eigenvalues correspond to idiosyncratic variation.

Population projection and the common component

Let $P_{n,q}^x(\theta)$ denote the matrix collecting the q eigenvectors corresponding to the largest q eigenvalues of $\Sigma_n^x(\theta)$.

The rank- q projection matrix

$$K_n^x(\theta) = P_{n,q}^x(\theta) P_{n,q}^x(\theta)^\dagger.$$

The corresponding linear filter coefficients (in the lag domain) are obtained via inverse Fourier transform:

$$K_n^x(h) = \frac{1}{8\pi^3} \int_{[-\pi,\pi]^3} K_n^x(\theta) e^{i\langle h,\theta \rangle} d\theta.$$

Then the population common component can be expressed as a filtered version of the observed field:

$$\chi_{\ell\varsigma} = \sum_{m=1}^n \sum_{h \in \mathbb{Z}^3} K_{n,\ell m}^x(h) x_{m,\varsigma-h}. \quad (4)$$

Methodology

Estimation: kernel-smoothed estimator of the spectral density matrix

In finite samples, $\Sigma_n^x(\theta)$ is unknown and must be estimated.

$$\hat{\Sigma}_n^x(\theta) = \frac{1}{S_1 S_2 T} \sum_{s_1} \sum_{s_2} x_{n,s_1} x_{n,s_2}^\top K_1\left(\frac{s_{11} - s_{21}}{B_{S_1}}\right) K_2\left(\frac{s_{12} - s_{22}}{B_{S_2}}\right) K_3\left(\frac{t_1 - t_2}{B_T}\right) e^{-i\langle s_1 - s_2, \theta \rangle}. \quad (5)$$

where K_1, K_2, K_3 are kernel functions and (B_{S_1}, B_{S_2}, B_T) are bandwidth parameters.

Note: in practice, (5) is evaluated on a discrete frequency grid $\{\theta_h\}_{h=1}^{S_1 S_2 T}$ corresponding to spatio-temporal Fourier frequencies.

Algorithm 1: Estimation of the common component

[1]

Data $\{x_{\ell\varsigma}\}$, $\ell = 1, \dots, n$, $\varsigma = (s_1, s_2, t)^\top$; estimated number of factors $\hat{q}^{(n)}$; kernels K_1, K_2, K_3 ; bandwidths (B_{S_1}, B_{S_2}, B_T) ; truncation parameters (M_{S_1}, M_{S_2}, M_T) .

Estimated common component $\{\hat{\chi}_{\ell\varsigma}^{(n)}\}$.

Compute $\hat{\Sigma}_n^x(\theta_h)$ on the frequency grid using (5).

for each frequency θ_h **do**

 Compute the leading $\hat{q}^{(n)}$ eigenvectors of $\hat{\Sigma}_n^x(\theta_h)$.

 Construct $\hat{K}_n^x(\theta_h) = \hat{P}_{n,\hat{q}}^x(\theta_h) \hat{P}_{n,\hat{q}}^x(\theta_h)^\dagger$.

end for

Obtain truncated filter coefficients $\hat{K}_n^x(h)$ by inverse Fourier transform.

Compute $\hat{\chi}_{\ell\varsigma}^{(n)}$ using

$$\hat{\chi}_{\ell\varsigma}^{(n)} = \sum_{m=1}^n \sum_{|h_1| \leq M_{S_1}} \sum_{|h_2| \leq M_{S_2}} \sum_{|h_3| \leq M_T} \hat{K}_{n,\ell m}^x(h) x_{m,\varsigma-h}.$$

return $\{\hat{\chi}_{\ell\varsigma}^{(n)}\}$.

Determining the number of factors: Information criterion

$$IC^{(n)}(k) = \log \left[\frac{1}{n} \sum_{j=k+1}^n \left(\frac{1}{8\pi^3} \int_{[-\pi, \pi]^3} \hat{\lambda}_{nj}^x(\theta) d\theta \right) \right] + k p(n, S_1, S_2, T), \quad (6)$$

where $\hat{\lambda}_{nj}^x(\theta)$ denote the j -th largest eigenvalue of $\hat{\Sigma}_n^x(\theta)$, for $k = 0, \dots, q_{\max}$.

The estimated factor number is

$$\hat{q}^{(n)} = \arg \min_{0 \leq k \leq q_{\max}} IC^{(n)}(k).$$

Algorithm 2: Selection of the number of factors

Algorithm 1 Algorithm 2: Selection of the number of factors $\hat{q}^{(n)}$

[1]

Data $\{x_{\ell k}\}$; maximum number of factors $q_{\max}$; subsample sizes $n_1 < \dots < n_J = n$; constants $c_1, \dots, c_L$; kernels K_1, K_2, K_3 ; bandwidths (B_{S_1}, B_{S_2}, B_T) .

$\hat{q}^{(n)}$.

Choose a random permutation of the n cross-sectional units.

for $\ell = 1, \dots, L$ **do**

for $j = 1, \dots, J$ **do**

 Compute $\hat{\Sigma}_{n_j}^x(\theta_h)$ on the grid using (5).

 Compute the $q_{\max}$ largest eigenvalues of $\hat{\Sigma}_{n_j}^x(\theta_h)$.

 Evaluate $IC^{(n_j)}(k)$ in (6) for $k = 0, \dots, q_{\max}$ using penalty constant c_ℓ .

 Set $\hat{q}_{c_\ell}^{(n_j)} = \arg \min_{0 \leq k \leq q_{\max}} IC^{(n_j)}(k)$.

end for

 Compute a stability/variability score S_{c_ℓ} from $\{\hat{q}_{c_\ell}^{(n_j)}\}_{j=1}^J$.

end for

Choose $\hat{c}$ from the second stability region in the plot $c \mapsto c S_c$.

return $\hat{q}^{(n)} = \hat{q}_{\hat{c}}^{(n)}$.

Application: WeatherBench2 / ERA5

Data construction

- The sample: January 2010 at 6-hourly frequency and a European spatial subdomain.
- Four variables defined on pressure levels: temperature, geopotential, u_component_of_wind, v_component_of_wind.
- Each variable yields a four-dimensional block of size

$$(8, 23, 27, 124),$$

corresponding to (level, latitude, longitude, time).

•

$$X_{\text{raw}} \in \mathbb{R}^{32 \times 23 \times 27 \times 124}, \quad n = 4 \times 8 = 32.$$

- Observed on a 23×27 spatial grid over $T = 124$ time points.

Standardization and tuning choices

For each ℓ , compute global space–time mean μ_ℓ and standard deviation σ_ℓ , and define

$$x_{\ell,s_1,s_2,t}^{(sc)} = \frac{x_{\ell,s_1,s_2,t} - \mu_\ell}{\sigma_\ell}.$$

Bandwidths:

$$B_{S_1} = B_{S_2} = B_T = 2.$$

Truncation:

$$|h_1| \leq M_{S_1}, \quad |h_2| \leq M_{S_2}, \quad |h_3| \leq M_T, \quad M_{S_1} = M_{S_2} = M_T = 1.$$

Sensitivity checks with slightly larger bandwidths and truncation windows yield qualitatively similar results.

Results: Estimated number of factors $\hat{q}$

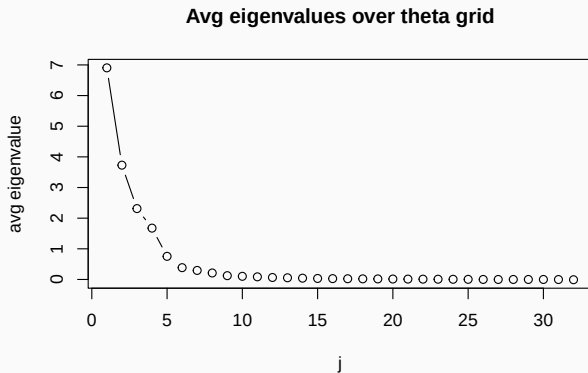


Figure 3: Frequency-averaged eigenvalues of the kernel-smoothed spatio-temporal spectral density estimator. The sharp drop after the first eigenvalue supports the choice $\hat{q} = 1$.

Results

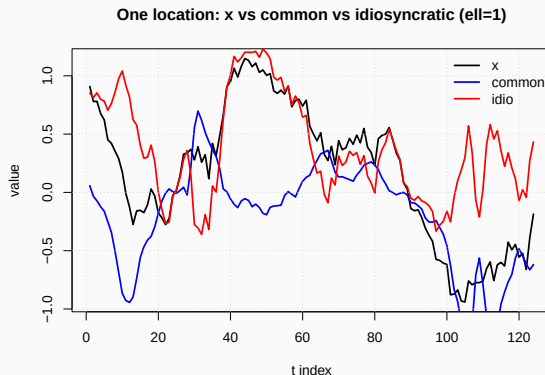


Figure 4: Time-series decomposition at one central spatial location.

Explained variance of the common component ($EV = 0.8522365$).

Thank you